INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, KALYANI

Pharmacy Inventory & Expiry Management System

Updated Project Progress Report

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Status of work completed to date and plan for completion

Contents

Abstract

1. Project Objective and Scope

2. Technology and Architecture

3. Work Completed to Date

4. Current Milestone Status

5. Work Remaining

6. Proposed Completion Plan

7. Conclusion

Declaration of Data Scope

Abstract

This report presents the current development status of the Pharmacy Inventory & Expiry Management System. The system is being developed as a full-stack web application to reduce medicine wastage, prevent stock-outs, improve batch-level traceability, and support pharmacy staff in daily inventory operations. The implemented solution combines a React and Vite frontend, a FastAPI and SQLAlchemy backend, a relational database, role-based access control, expiry monitoring, FEFO dispensing, inventory forecasting, procurement workflows, sales records, and exportable reports.

The core application workflow is now implemented and connected end to end for the main demonstration scenarios. The remaining work is primarily focused on systematic testing, production deployment hardening, usability refinement, documentation, and final evaluation using representative test data.

1. Project Objective and Scope

Manual registers and spreadsheets make it difficult to identify medicines that are approaching expiry, select the correct batch for dispensing, maintain accurate stock balances, and coordinate procurement. These limitations can lead to avoidable wastage, stock-outs, delayed reporting, and weak audit trails.

Objectives

• Maintain a structured catalogue of medicines, suppliers, batches, and pharmacy workspaces.

• Track stock quantities and expiry dates at batch level.

• Apply the First-Expiry-First-Out (FEFO) rule during dispensing.

• Generate low-stock, near-expiry, and demand-based replenishment recommendations.

• Record sales, purchases, returns, disposal, and operational audit events.

• Provide role-aware access for pharmacy administrators and pharmacists.

• Produce downloadable CSV and PDF reports for operational review.

2. Technology and Architecture

Layer	Technologies and implementation
Frontend	React 18, TypeScript, Vite, Chart.js, responsive CSS interface
Backend	Python, FastAPI, Pydantic schemas, SQLAlchemy ORM
Database	SQLite for local demonstration and PostgreSQL-ready configuration
Forecasting	XGBoost-based demand forecasting with a deterministic fallback
Reporting	CSV exports and PDF report endpoints
Communication	SMTP configuration for expiry and operational notifications
Deployment	Local development workflow with Docker Compose support for PostgreSQL

The application is organized as a multi-pharmacy platform. Records are linked to the signed-in pharmacy workspace so that medicines, batches, procurement, sales, and reports remain isolated between pharmacies. The backend creates and updates the required database structures at startup and seeds synthetic demonstration data for local evaluation.

3. Work Completed to Date

Foundation and Access Control

• Implemented a FastAPI backend with structured API routes, database sessions, ORM entities, validation schemas, and application configuration.

• Implemented pharmacy registration and multi-tenant workspace initialization.

• Implemented login and role-aware access for administrators and pharmacists.

• Added pharmacy-scoped access checks for operational records.

• Added a health endpoint and startup database compatibility handling.

Medicine Catalogue and Inventory

• Implemented medicine catalogue creation, editing, search, and category/reference data.

• Implemented batch-level stock records with batch number, quantity, expiry date, pricing, and disposal information.

• Added reorder levels, ideal stock values, low-stock indicators, and inventory summary metrics.

• Added CSV catalogue import with create-or-update behaviour by SKU.

• Added supplier records, purchase orders, and procurement requests.

Expiry, FEFO, and Waste Accountability

• Implemented expiry risk zones for safe, approaching, critical, and expired stock.

• Implemented near-expiry monitoring and dashboard alerts.

• Implemented FEFO-assisted dispensing so the earliest suitable batch is selected first.

• Prevented expired batches from normal sale or dispensing workflows.

• Implemented supplier return and licensed biomedical-waste disposal records with disposal method, reference, date, and audit information.

• Added company recovery records for eligible expired stock, including recovery values, transferred quantities, and handover references.

Sales, Transactions, Forecasting, and Reporting

• Implemented medicine dispensing with destination department and internal reference details.

• Implemented itemised sales invoices and sales ledger information.

• Implemented transaction and consumption history.

• Added 30-day, 60-day, and 90-day demand forecast views with an XGBoost model and deterministic fallback.

• Implemented downloadable CSV and PDF inventory reporting surfaces.

• Added SMTP configuration and an email delivery check for expiry, stock, and supplier-return notifications.

4. Current Milestone Status

Milestone	Status	Evidence in the implementation
Project structure and backend foundation	Completed	FastAPI application, SQLAlchemy models, configuration, database session, and startup lifecycle.
Frontend application shell	Completed	React/Vite application with operational pages and reusable UI patterns.
Authentication and pharmacy workspaces	Completed	Login, administrator/pharmacist roles, pharmacy registration, and workspace scoping.
Medicine and batch inventory	Completed	CRUD workflows, batch expiry tracking, thresholds, pricing, and stock summaries.
Expiry and waste management	Completed	Risk zones, quarantine, supplier returns, recovery records, disposal references, and audit trail.
FEFO dispensing and sales	Completed	Earliest-expiry batch selection, dispensing records, invoices, and sales totals.
Procurement and suppliers	Completed	Suppliers, purchase orders, procurement requests, approval workflow, and status tracking.
Forecasting and alerts	Completed	Forecast views, reorder recommendations, low-stock and near-expiry alerts.
Exports and reporting	Completed	CSV/PDF reporting endpoints and reports page.
Automated testing and test evidence	In progress	Functional scenarios still need to be executed systematically and documented.
Production deployment and security hardening	Remaining	Final environment configuration, secret review, deployment rehearsal, and access review remain.
Final documentation and presentation	In progress	Technical documentation and final demonstration material remain to be consolidated.

5. Work Remaining

Testing and Validation

• Create a repeatable test dataset covering normal, low-stock, near-expiry, expired, returned, disposed, and recovered batches.

• Test all role permissions and confirm that one pharmacy cannot read or modify another pharmacy's records.

• Test FEFO selection with multiple batches and partial quantities.

• Test sales, purchases, returns, disposal, and recovery calculations against manually verified expected values.

• Test CSV import validation, duplicate SKUs, malformed rows, and empty-file behaviour.

• Test forecast behaviour with sufficient, insufficient, and unusual consumption history.

• Test PDF/CSV report contents and download behaviour.

• Record test cases, screenshots, defects, and fixes for the final evaluation chapter.

Deployment, Security, Usability, and Documentation

• Validate PostgreSQL deployment using environment variables rather than local demonstration defaults.

• Review authentication, password handling, CORS, SMTP, and API error responses.

• Confirm that real credentials and private configuration values are not included in the repository.

• Add database backup and restore instructions and perform one recovery rehearsal.

• Conduct a usability review, refine messages and empty states, and complete API, installation, and user-operation documentation.

• Prepare the final presentation, demonstration script, and individual contribution summary.

6. Proposed Completion Plan

Phase	Target	Activities
Validation	Week 1	Execute functional, role, data-integrity, import, report, and forecast tests.
Hardening	Week 2	Fix defects, review configuration and secrets, validate PostgreSQL deployment, and complete backup rehearsal.
Usability	Week 3	Refine interface based on walkthroughs, capture screenshots, and finalize user instructions.
Submission	Week 4	Complete final report, presentation, contribution summary, and final demonstration rehearsal.

7. Conclusion

The project has progressed from the initial interface and conceptual roadmap to a connected pharmacy-management application with the main operational workflows implemented. The strongest completed areas are batch-level inventory, expiry accountability, FEFO dispensing, multi-pharmacy isolation, procurement, sales, forecasting, and reporting.

The project is not yet considered final until the remaining validation, deployment, security review, usability review, and documentation activities have been completed. The next immediate priority is to run and document an end-to-end test cycle using controlled synthetic data, followed by deployment hardening and preparation of the final demonstration.

Declaration of Data Scope

The demonstration catalogue, suppliers, prices, stock quantities, and consumption history are synthetic project data. They are intended for software demonstration and testing only and do not represent real patient records, licensed stock, or a guarantee of regulatory approval. Any operational deployment must use verified local medicine, supplier, licensing, and disposal information.

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